

aucmat: matrix-first ROC analysis for high-dimensional biomarker screening in R

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Abstract

Summary: The area under the receiver operating characteristic curve (AUC) is the standard metric for evaluating continuous biomarkers against binary outcomes. When screening thousands of molecular biomarkers, researchers need matrix-wide AUC estimation with proper inference, multiplicity adjustment, and stability assessment. We present aucmat, an R package providing a matrix-first architecture: `aucmat(X, y)` computes direction-preserving AUCs with DeLong or stratified-bootstrap confidence intervals for every column of a numeric matrix in a single call. The package supports Benjamini-Hochberg multiplicity adjustment, three hypothesis-testing frameworks for paired comparisons (superiority, non-inferiority, equivalence via TOST), an omnibus Wald test for equality of multiple correlated AUCs via the joint DeLong covariance matrix, and bootstrap rank-stability analysis. Cross-validated screening eliminates same-data optimism. A two-stage Hurdle-AUC model handles zero-inflated data (scRNA-seq, microbiome) where standard AUC degrades to near chance. Eight simulation generators and a validation harness produce correlated biomarkers with controlled properties under parametrized correlation structures with built-in calibration diagnostics. The package provides over a dozen visualization functions including ranked discrimination plots, volcano plots, forest plots, ROC curves with confidence ribbons, precision-recall curves, and Hurdle-AUC diagnostics. On a typical proteomics screen, aucmat avoids the per-column looping required by single-biomarker tools and provides complete inference in a single function call. The package is freely available under the MIT license at <https://github.com/vanhungtran/aucmat>. **Availability and Implementation:** R package (R $\geq$ 4.0.0), MIT license. Source: <https://github.com/vanhungtran/aucmat>. Documentation: <https://vanhungtran.github.io/aucmat>. Dependencies: pROC ($\geq$ 1.18), mvtnorm, ggplot2, rlang. **Contact:** lucas.tran@ck-care.ch

Introduction

Receiver operating characteristic (ROC) analysis is the dominant framework for evaluating continuous biomarkers against binary outcomes in diagnostic medicine, prognostic modeling, and molecular screening [Hanley and McNeil, 1982, DeLong et al., 1988, Pepe, 2004]. The area under the ROC curve (AUC) summarizes the probability that a randomly chosen positive subject has a higher biomarker value than a randomly chosen negative subject. Its interpretability – 0.5 for chance, 1.0 for perfect separation – has made it the default metric in omics-scale biomarker discovery [Robin et al., 2011].

Modern molecular profiling technologies routinely measure thousands of features on a few hundred subjects. A proteomics panel may contain 5,000 proteins; a metabolomics screen 2,000 metabolites; a transcriptomics experiment 20,000 genes. The core statistical task – “which biomarkers discriminate cases from controls?” – requires computing and testing an AUC for each feature, adjusting for multiplicity, comparing top candidates, and assessing the stability of the rankings under sampling variability. Despite decades of methodological development, no widely-used R package integrates these steps into a single, reproducible workflow.

The existing R ecosystem provides excellent point solutions. **pROC** [Robin et al., 2011] is the gold standard for single-biomarker ROC inference: it computes DeLong confidence intervals using the Sun & Xu (2014) $O(N \log N)$ algorithm [Sun and Xu, 2014], tests paired or unpaired AUC differences via `roc.test()`, handles partial AUC with bootstrap inference, provides binormal and density-smoothed ROC curves, computes covariance between paired AUCs via `cov()`, and supports multiclass AUC via Hand & Till [Hand and Till, 2001]. However, pROC operates one biomarker at a time – screening a matrix of 10,000 features requires

an explicit per-column loop. **ROCR** [Sing et al., 2005] offers flexible visualization with 28 performance measures and cross-validation averaging but provides no statistical inference (no confidence intervals, no hypothesis tests for AUC differences). **precrec** [Saito and Rehmsmeier, 2017] adds precision-recall curves and fast C++-based AUC computation, with confidence bands for multiple test sets and functions for partial AUC (`part()`, `pauc()`) and AUC confidence intervals (`auc_ci()`), but provides no DeLong inference, no multiplicity adjustment, and no paired comparison tests. **colAUC** from caTools [Tuszynski and Dietze, 2024] computes column-wise AUCs efficiently via the Wilcoxon rank-sum test and supports multiclass AUC via column means – but provides no inference whatsoever (no standard errors, confidence intervals, or p-values). **dtComb** [Yerlitas et al., 2025] provides 142 methods for combining two diagnostic markers into a composite score, with 30+ cutoff optimization methods, resampling, and caret integration – but is limited to exactly two biomarkers and does not perform matrix-wide screening. **cancerclass** [Budczies et al., 2024] builds validated classifiers from high-dimensional molecular data using nearest-centroid prediction with multiple random validation, but does not offer DeLong-based screening or bootstrap stability analysis.

`aucmat` fills this gap. Its design philosophy is *matrix-first*: the primary function `aucmat(X, y)` accepts a numeric matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a binary outcome vector $\mathbf{y} \in \{0, 1\}^n$, and returns a complete screening table with point estimates, standard errors, confidence intervals, raw and multiplicity-adjusted p-values, and diagnostic metadata for every biomarker in a single call. Downstream functions for comparison, stability, cross-validation, panel construction, and power analysis consume the same S3 object, preserving outcome encoding and inferential settings throughout.

Methods

Direction-preserving AUC

For biomarker X and binary outcome $Y \in \{0, 1\}$, the AUC equals the Mann-Whitney U probability: $\text{AUC} = P(X_i > X_j \mid Y_i = 1, Y_j = 0)$ [Hanley and McNeil, 1982]. Most ROC software reports $\max(\text{AUC}, 1 - \text{AUC})$, silently reversing the biomarker to force the reported value above 0.5. `aucmat` deliberately preserves the observed direction, reporting three values per biomarker: AUC_{raw} (the Mann-Whitney estimate on the original numeric scale, which may be below 0.5), $\text{AUC}_{\text{strength}} = 0.5 + |\text{AUC}_{\text{raw}} - 0.5|$ (for ranking), and `effect_direction` (`higher_in_positive` or `lower_in_positive`). This distinction matters biologically: a tumor suppressor gene with $\text{AUC}_{\text{raw}} = 0.15$ discriminates just as strongly as an oncogene with $\text{AUC}_{\text{raw}} = 0.85$, but with the opposite biological interpretation.

DeLong variance and joint covariance

The DeLong variance estimator [DeLong et al., 1988, Sun and Xu, 2014] uses placement values $V_{10}(X_i) = n_0^{-1} \sum_{j: Y_j=0} [\mathbf{1}(X_j < X_i) + 0.5 \cdot \mathbf{1}(X_j = X_i)]$ for positive subjects and $V_{01}(X_j) = n_1^{-1} \sum_{i: Y_i=1} [\mathbf{1}(X_i > X_j) + 0.5 \cdot \mathbf{1}(X_i = X_j)]$ for negative subjects. Let $S_{10} = (n_1 - 1)^{-1} \sum_{i \in \mathcal{I}_1} (V_{10}(X_i) - \bar{V}_{10})^2$ and $S_{01} = (n_0 - 1)^{-1} \sum_{j \in \mathcal{I}_0} (V_{01}(X_j) - \bar{V}_{01})^2$ be the sample variances of the placement values. The DeLong variance estimator is $\widehat{\text{Var}}(\widehat{\text{AUC}}) = S_{10}/n_1 + S_{01}/n_0$. For p biomarkers on common subjects, the $p \times p$ joint covariance matrix $\widehat{\Sigma} = n_1^{-1} \widehat{\text{Cov}}(\mathbf{V}_{10}) + n_0^{-1} \widehat{\text{Cov}}(\mathbf{V}_{01})$ supports an omnibus Wald test of $H_0 : \text{AUC}_1 = \dots = \text{AUC}_p$. Let $\mathbf{C}$ be a $(p - 1) \times p$ contrast matrix (default: comparing biomarkers 2, $\dots$, p against biomarker 1). The Wald statistic $W = (\mathbf{C}\hat{\theta})^\top (\mathbf{C}\widehat{\Sigma}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\theta}) \sim \chi_r^2$ uses the effective rank r of the contrast covariance matrix, determined via eigenvalue tolerance. A generalized inverse is used on the stable subspace only; rank-zero or numerically unstable cases return `status = "non_testable_singular"`. For $p = 2$, W equals the squared paired-DeLong z -statistic within numerical tolerance, serving as an internal consistency check.

Multiplicity adjustment and stability

Screening p biomarkers produces p hypothesis tests. `aucmat` applies Benjamini-Hochberg [Benjamini and Hochberg, 1995], Holm, or Bonferroni correction via `stats::p.adjust()`, reporting adjusted p-values as q -values. For equivalence (TOST) [Schuirmann, 1987], the reported $p_{\text{equiv}} = \max(p_{\text{lower}}, p_{\text{upper}})$ and multiplicity adjustment applies to this combined value.

Bootstrap rank-stability analysis (`auc_stability()`) resamples the data B times with replacement (stratified by outcome), recomputes the full screening for each replicate, and reports median/IQR rank distributions, top- k selection probabilities, and pairwise co-selection frequencies – quantifying how reproducible the ranking is under sampling variability [Efron and Tibshirani, 1994].

Hurdle-AUC for zero-inflated data

Standard AUC assumes continuous biomarkers following a shifted normal distribution per class. This assumption breaks on zero-inflated data (scRNA-seq, microbiome, detection-limited assays) where 60%+ of values are exactly zero. Under the standard model, zeros would need to arise from the tail of a normal distribution below zero – biologically implausible. The Hurdle-AUC uses a two-stage model:

Stage 1 – Zero hurdle: $P(X_j = 0 | Y) = \text{logistic}(\beta_0 + \beta_1 Y)$ models the binary expression decision via logistic regression.

Stage 2 – Magnitude: Standard AUC on the non-zero values captures magnitude discrimination among expressed observations.

The composite score S_i for observation i is: $S_i = \hat{p}_i$ if $X_{ij} = 0$, and $S_i = 0.5 \cdot \hat{p}_i + 0.5 \cdot \tilde{x}_{ij}$ if $X_{ij} > 0$, where $\hat{p}_i$ is the predicted probability of being non-zero from Stage 1, and $\tilde{x}_{ij}$ is the normalized biomarker value.

Simulation confirms recovery from standard AUC ≈ 0.53 to Hurdle-AUC ≈ 0.90 on zero-inflated biomarkers by acknowledging that “expression on/off” and “expression level when on” are separate biological processes.

Multivariate simulation

Under the equal-variance binormal model, $X_k | Y \sim \mathcal{N}(\mu_{kY}, \sigma^2)$ with $\delta_k = (\mu_{1k} - \mu_{0k})/\sigma = \sqrt{2} \cdot \Phi^{-1}(\text{AUC}_k)$. For p biomarkers with correlation matrix $\mathbf{R}$, the class-conditional distribution is $\mathbf{X} | Y \sim \mathcal{N}_p(\boldsymbol{\mu}_Y, \mathbf{R})$, where $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_1$ are centred so the n -weighted grand mean is zero. `simulate_auc_matrix()` generates data under this model with four parametrized correlation structures (user, exchangeable, AR(1), block) and feasibility diagnostics with optional nearest positive-definite projection.

`simulate_auc_copula()` uses a two-phase approach: Phase 1 applies a Gaussian copula (rank-transform to uniform, probit to normal scores) that preserves the target correlation structure; Phase 2 applies iterative mean perturbation with decaying step size to fine-tune each biomarker’s AUC. This separates correlation control (copula) from discriminative signal (perturbation), avoiding the AUC-correlation tradeoff inherent in sequential binormal methods.

`simulate_hurdle_auc()` implements the two-stage hurdle model: Bernoulli zeros with class-specific rates, and log-normal magnitudes for expressed values with binormal-calibrated mean separation, clipped to $[0, \infty)$.

`validate_simulation()` repeats a specification across independent draws and reports bias, RMSE, Monte Carlo standard error, and target-interval hit rates for both AUCs and pairwise correlations – a single draw is never evidence that a simulator is calibrated.

Implementation

`aucmat` is implemented in pure R (~5,700 lines across 35 source files) with minimized dependencies. The primary function `aucmat()` performs six steps in sequence: (1) input validation (matrix dimensions, column names, outcome levels, infinite values); (2) outcome normalization via `.normalize_binary_y()` (accepting logical, numeric 0/1, factor, or character outcomes); (3) matrix-wide AUC computation via the Mann-Whitney rank formulation; (4) DeLong or bootstrap inference via `.apply_inference()` with configurable alternative hypothesis (`two.sided`, `greater`, `less`); (5) multiplicity adjustment via `.adjust_pvalues()`; and (6) result assembly with ranking by discrimination strength and feature-level warnings for small class counts or high missingness.

All primary functions return S3 objects with `print()`, `summary()`, `plot()`, and (where appropriate) `as.data.frame()` and `subset()` methods. The package stores outcome encoding, positive-class design-

nation, and sample masks in the returned object, eliminating the need for downstream functions to re-infer factor direction or re-derive class counts.

Structured error classes (inheriting from `error`) allow programmatic handling of common failures: `aucmat_invalid_outcome`, `aucmat_invalid_correlation`, `aucmat_infeasible_targets`, `aucmat_insufficient_sample`, and `aucmat_resampling_failure`. An RNG-safety mechanism (`with_seed()`) saves the global `.Random.seed`, executes the computation, and restores the previous state on exit, including when `.Random.seed` was previously absent and when an error occurs.

The package imports pROC (ROC curve objects and smoothed ROC), mvtnorm (multivariate normal generation), ggplot2 (visualization), and rlang (tidy evaluation). Optional Suggests include GGally (pairs plots for simulated data), glmnet (penalized panel models), matrixStats (fast column-wise ranks), knitr and rmarkdown (vignette building), and ggrepel (non-overlapping plot labels). The complete source, documentation, and four vignettes are available at <https://vanhungtran.github.io/aucmat>.

Usage

Basic screening

```
library(aucmat)

sim <- simulate_auc_matrix(
  n = 500, prevalence = 0.3,
  target_aucs = c(0.9, 0.8, 0.7, 0.65),
  correlation = 0.3, structure = "exchangeable"
)
X <- as.matrix(sim$data[, 1:4])
y <- sim$data$truth

fit <- aucmat(X, y, ci = "delong", adjust = "BH")
print(fit)
summary(fit)
plot_auc_rank(fit, n_label = 4)
```

Paired comparisons with hypothesis testing

```
# Two-sided superiority (default: H0: delta = 0)
compare_auc(fit, X, y, reference = "X1")

# Non-inferiority: H0: AUC_a <= AUC_b - margin
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "noninferiority", margin = 0.05)

# Equivalence (TOST): H0: |AUC_a - AUC_b| >= margin
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "equivalence", margin = 0.15, adjust = "BH")

# Omnibus Wald test: H0: AUC_1 = AUC_2 = ... = AUC_p
compare_auc_global(fit, X, y, biomarkers = paste0("X", 1:4))

# Single-biomarker DeLong test against chance
roc_test(X[, 1], y)
```

Hurdle-AUC for zero-inflated data

```
# scRNA-seq: 55% zeros in controls, 25% in cases
sim <- simulate_hurdle_auc(
  n = 400, prevalence = 0.3,
  target_hurdle_aucs = c(0.85, 0.72),
  zero_rate_neg = c(0.55, 0.80),
  zero_rate_pos = c(0.25, 0.70)
)
fit_hur <- hurdle_auc(as.matrix(sim$data[, 1:2]), sim$data$truth)
print(fit_hur)
# Standard AUC ~0.53 recovers to Hurdle AUC ~0.90
```

Cross-validation, panel construction, power analysis

```
# Cross-validated AUC (honest out-of-fold estimates)
cv <- cv_aucmat(X, y, n_folds = 5, seed = 42)

# Combine top biomarkers into a penalized panel
panel <- fit_auc_panel(X[, head(cv$results$biomarker, 4)], y,
  method = "ridge", n_folds = 5)

# Sample size for 80% power to detect AUC = 0.70
power_auc_matrix(
  target_aucs = 0.70, power = 0.80,
  prevalence = 0.3, alpha = 0.05
)
```

Benchmarking

Screening throughput. On a desktop workstation (AMD Ryzen 3.5 GHz, 32 GB RAM, R 4.5.1, Windows 11), aucmat screens 100, 1,000, and 10,000 biomarkers ($n = 500$, DeLong inference) in 0.54, 5.02, and 59.76 seconds respectively. The equivalent per-column pROC loop (`pROC::roc()` then `pROC::ci.auc()` with `method = "delong"`) on the same data requires 0.68, 6.42, and an estimated 64 seconds – a speedup of approximately 1.1–1.3 $\times$ at moderate p , increasing at larger p as aucmat’s fixed per-call overhead is amortized over more columns. The speed advantage comes from avoiding repeated outcome encoding and from column-wise rank computation. At small p (< 100), the difference is negligible; for wide matrices ($p > 1,000$), the per-column overhead of `pROC::roc()` and `pROC::ci.auc()` becomes meaningful.

Simulation calibration. Across 27 simulation conditions ($n = 100, 300, 500$; prevalence averaged over 0.2, 0.3, 0.5; target AUC = 0.65, 0.75, 0.85; 200 replicates each; averaged over exchangeable and AR(1) correlation structures with $\rho = 0$ and $\rho = 0.3$), DeLong 95% confidence interval coverage ranged from 0.898 (at $n = 100$, AUC = 0.65) to 0.957 (at $n = 500$, AUC = 0.65), converging to the nominal 0.95 as sample size increased. Power exceeded 0.85 for AUC ≥ 0.65 at $n = 100$ and reached 1.0 for AUC ≥ 0.75 at $n \geq 300$. The coverage is slightly below nominal for small n and extreme AUC values, consistent with the known finite-sample behavior of the DeLong variance estimator.

Table 1: Empirical coverage of DeLong 95% confidence intervals, averaged across prevalence (0.2, 0.3, 0.5) and correlation structures (exchangeable, AR1) with 200 replicates each.

	AUC	n=100	n=300	n=500
1	0.65	0.898	0.937	0.957
4	0.75	0.915	0.943	0.933
7	0.85	0.883	0.923	0.905

Comparison with existing R packages

Table ?? compares aucmat (v0.2.0) with five established R packages for ROC analysis and biomarker evaluation. Features are verified against each package’s CRAN documentation and published literature as of July 2026.

Table 2: Feature comparison of R packages for ROC analysis and biomarker screening. \texttt{+} = supported; \texttt{-} = not supported or requires manual implementation. Verified against CRAN documentation as of July 2026. pROC supports unpaired comparisons via \texttt{roc.test(paired=FALSE)}; aucmat requires common subjects for all comparisons.

Feature	aucmat	pROC	precrec	ROCR	colAUC	dtComb
Matrix-first screening (single call)	+	-	-	-	+	-
Direction-preserving AUC (<0.5 reported)	+	-	-	-	-	-
DeLong confidence intervals	+	+	-	-	-	-
Stratified bootstrap CI	+	+	-	-	-	+
Multiplicity adjustment (BH/Holm/Bonf)	+	-	-	-	-	-
One-sided alternative tests	+	+	-	-	-	-
Paired biomarker comparisons	+	+	-	-	-	+
Non-inferiority / equivalence (TOST)	+	-	-	-	-	-
Omnibus Wald test (3+ correlated AUCs)	+	-	-	-	-	-
Bootstrap rank-stability analysis	+	-	-	-	-	-
Partial AUC	+	+	+	-	-	-
Multivariate simulation engine	+	-	-	-	-	-
Simulation calibration diagnostics	+	-	-	-	-	-
Cross-validated AUC screening	+	-	-	-	-	+
Penalized panel models (ridge/lasso)	+	-	-	-	-	+
Hurdle-AUC (zero-inflated data)	+	-	-	-	-	-
Precision-Recall curves	+	-	+	-	-	-
Power / sample size computation	+	+	-	-	-	-
Multiclass AUC (Hand-Till)	+	+	-	-	+	-
ROC smoothing (binormal/density)	+	+	-	-	-	-
Group-stratified screening	+	-	-	-	-	-

What aucmat uniquely provides. Five capabilities are, to our knowledge, not available in any other single R package: (i) matrix-first screening with direction-preserving AUC and multiplicity adjustment in one call; (ii) non-inferiority and equivalence (TOST) tests for paired AUC comparisons; (iii) an omnibus Wald test for three or more correlated AUCs via the joint DeLong covariance matrix; (iv) bootstrap rank-stability analysis with top- k selection probabilities and pairwise co-selection frequencies; and (v) a two-stage Hurdle-AUC model for zero-inflated data that recovers discriminative signal where standard AUC fails.

When to use what. For a single biomarker with smoothed ROC curves, pROC remains the gold standard, offering binormal and density smoothing, partial AUC with bootstrap inference, and Venkatraman’s permutation test for whole-curve comparison [Venkatraman and Begg, 2000]. For speed on extremely wide matrices (>100k columns), colAUC’s Wilcoxon-based column-wise computation is recommended, though it provides no inference. For precision-recall analysis of highly imbalanced data, precrec’s C++ backend offers fast and accurate PR curves with confidence bands for multiple test sets. For combining two biomarkers into an optimal composite score, dtComb’s library of 142 methods covers virtually all published linear, non-linear, mathematical, and machine-learning combination approaches. For end-to-end screening of hundreds to tens of thousands of biomarkers with full statistical inference, multiplicity adjustment, comparison testing, stability assessment, validation, and study planning in a single coherent workflow, aucmat is the recommended tool.

Discussion

aucmat provides a unified, statistically principled framework for biomarker screening against binary outcomes in R. Its matrix-first design reduces the code burden for high-dimensional ROC analysis from dozens of bespoke scripting steps to a handful of function calls with consistent S3 interfaces. Cross-validated screening and penalized panel models address same-data optimism, while power analysis tools support prospective study design.

Limitations. The current version (0.2.0) is restricted to binary outcomes for the primary screening workflow (multiclass is available experimentally via `aucmat_multiclass()`). The DeLong variance estimator requires common subjects for paired comparisons; unpaired comparisons between independent cohorts are not yet implemented (pROC provides this via `roc.test(paired=FALSE)`). Partial AUC computation is implemented as an internal engine (R/`partial-auc.R`) but not yet exposed through the primary `aucmat()` interface. The package targets in-memory dense matrices; sparse and on-disk backends are deferred. The DeLong variance estimator shows slightly liberal coverage at small sample sizes (coverage ≈ 0.90 at $n = 100$ for $AUC = 0.65$), consistent with its known finite-sample behavior; bootstrap intervals are recommended when $n < 100$ per class.

Future work. The development roadmap targets four major extensions: version 0.3 will add multivariable penalized panel models with honest nested cross-validation to prevent information leakage across preprocessing, feature selection, and tuning steps; version 0.4 will extend the framework to multiclass outcomes with one-versus-rest, one-versus-one, and Hand-Till estimators; version 0.5 will incorporate cumulative/dynamic time-dependent AUCs for censored survival outcomes [Heagerty et al., 2000, Blanche et al., 2013]; and version 0.6 will introduce sparse-matrix and on-disk backends for genomics-scale data exceeding available RAM.

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